Are Machine Learning Improvements Statistically Real? Dependence-Aware Inference for Reliable Model Comparison on Spatiotemporal Data

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Abstract

Claims that one machine learning model outperforms another usually rest on cross-validated scores and a test that assumes independent observations. On spatial and temporal data this assumption fails twice: random cross-validation leaks information between neighbouring observations, and the prediction errors of the compared models are themselves dependent. We study the second, less examined problem. We cast model comparison as inference on the mean of a dependent field of out-of-fold loss differentials with correlated fold-level training effects, and show that the paired t-test on fold scores underestimates the variance by exactly the average covariance between fold means. We propose DA-HAC, a dependence-aware test computed from a single blocked cross-validation run. It combines a product-kernel variance estimator with bandwidths calibrated on the empirical variogram of the loss differentials, a method-of-moments estimate of the training variance, and a Satterthwaite reference distribution. In simulations with exact control of the truth, including XGBoost learners whose expected performances are equal by construction, conventional procedures reject a true null hypothesis in up to **80%** of replicates, and random cross-validation halves the estimated gain of a genuinely better model under strong dependence. Blocked cross-validation with DA-HAC keeps the error rate near the nominal level under weak dependence and reduces it substantially otherwise, while diagnosing, through an effective sample size and a range ratio, the cases in which no reliable comparison is possible. On hourly weather, pedestrian counts and yield-monitor data, the conclusions of four of six comparisons depend on these choices. An open-source Python implementation is provided.

Keywords: Model comparison, Cross-validation, Spatio-temporal dependence, HAC variance estimation, Variogram, Statistical significance, Trustworthy machine learning

1 Introduction

Empirical machine learning advances one comparison at a time. A new gradient-boosting configuration is reported to beat a random forest, a deep architecture to beat gradient boosting, a feature set to beat another. Such claims rest on a small number of summary statistics (an average accuracy, an F1-score, a root mean squared error) computed by cross-validation, and, in the better cases, on a statistical test or a confidence interval for the difference. The test is meant to answer a simple question: *is the observed improvement larger than what the randomness of the data and of the validation procedure could produce on its own?*

The standard tools for that question [1–4] were developed for observations that are independent and identically distributed (i.i.d.). Much of the data on which machine learning is now used in practice is not. Meteorological, environmental, agricultural, epidemiological and mobility data are collected at locations and times that are close to one another, and nearby observations share unmeasured drivers. Two consequences are well documented. First, random K-fold cross-validation places near-duplicates of test observations in the training set, so that it measures interpolation rather than prediction at new places or times, and is optimistic when the latter is the target [5–8]. Second, and much less studied, the *errors* of a fitted model are themselves dependent, so that n test observations carry much less information than n independent ones.

The first problem has a well-known remedy: blocked (spatial, temporal or spatio-temporal) cross-validation [5, 9, 10]. The second problem is the subject of this paper. Blocking changes *what* is estimated; it does not tell us *how precisely* it is estimated. A practitioner who computes a blocked cross-validation score for two models and then applies a paired t-test, a bootstrap that resamples observations, or a test on fold-wise scores still relies on independence assumptions that the data violate. Whether an apparent improvement is "statistically real" then depends as much on these hidden assumptions as on the models.

Our starting point is the per-observation out-of-fold *loss differential* $d_i = \ell(y_i, \hat{f}_A(x_i)) - \ell(y_i, \hat{f}_B(x_i))$ between two models A and B. The mean of the d_i estimates the expected gain of A over B, and the comparison is a problem of inference on the mean of a dependent random field. Two sources of dependence must be handled at once. (i) Test-point dependence: the d_i inherit the spatial and temporal autocorrelation of the residuals of both models. (ii) Training dependence: all observations of a fold are predicted by the same fitted models, and the models of different folds are fitted on overlapping training sets. Existing procedures handle one source at most. Heteroskedasticity- and autocorrelation-consistent (HAC) variance estimators, as in the Diebold–Mariano test for forecasts [11] and its spatial extension [12], handle (i) for fixed predictors. The corrected resampled t-test of Nadeau and Bengio [2] handles (ii) for i.i.d. data.

Contributions.

The paper makes four contributions.

1. We formalise model comparison under spatio-temporal dependence as inference on the mean of a loss-differential field with a fold-level random effect, and we show (Proposition 1) that the widely used paired t-test on fold scores underestimates the variance by exactly the average covariance between fold means. This explains why it fails under random cross-validation even though it "looks" conservative with only $K-1$ degrees of freedom.
2. We propose *DA-HAC*, a dependence-aware HAC procedure for comparing two models evaluated by blocked cross-validation. It combines (a) a product-Bartlett kernel estimator of the test-sampling variance whose spatial and temporal bandwidths are *calibrated from the empirical variogram of the loss differentials themselves*; (b) a method-of-moments estimate of the training-induced fold variance, inflated by a Nadeau–Bengio factor; and (c) a Welch–Satterthwaite reference distribution whose degrees of freedom reflect the effective sample size. All three steps are computed from a single cross-validation run.
3. We assess the validity of conventional and dependence-aware procedures (paired t, i.i.d. bootstrap, corrected resampled t, block bootstrap, linear mixed model, Diebold–Mariano–HAC and DA-HAC) in two simulation studies with exact control of the truth: one in which loss-differential fields are generated directly, and one with gradient-boosting learners trained on spatio-temporal data in which two models have equal expected loss by an exchangeability argument.
4. We apply all procedures to three public data sets (hourly weather in New York, hourly pedestrian counts in Melbourne, and yield-monitor maize data in Argentina), including a "seed-null" diagnostic in which a model is compared with itself, and we derive practical recommendations. An open-source Python implementation is provided.

The emphasis of the paper is thus not on whether spatial or temporal validation is preferable, a question already addressed in the literature, but on the *validity of the inference* drawn from a given validation design. In the language of the call for this special issue, it is a contribution to measuring the trustworthiness of comparisons between machine learning models in real applications.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 sets up the problem and analyses conventional tests. Section 4 presents DA-HAC. Sections 5 and 6 report the simulation studies, Section 7 the case studies, and Section 8 discusses recommendations and limitations.

2 Related work

Statistical comparison of learning algorithms.

Dietterich [1] showed that the paired t-test on resampled hold-out scores has an inflated type I error rate, because training sets overlap, and proposed the 5×2 cross-validation test. Nadeau and Bengio [2] derived the corrected resampled t-test, which inflates the variance of repeated train/test differences by the factor $1/J + n_{\text{test}}/n_{\text{train}}$. Bengio and

Grandvalet [13] proved that no unbiased estimator of the variance of K-fold cross-validation exists, which is why every practical procedure relies on a working model or a heuristic. Bouckaert and Frank [14] studied the replicability of these tests, Demšar [3] and Benavoli et al. [4] treated comparisons over multiple data sets, frequentist and Bayesian. More recently, Bayle et al. [15] established central limit theorems for K-fold cross-validation under algorithmic stability, and Bates et al. [16] clarified that cross-validation estimates the average error of the learning *procedure* rather than that of the fitted model, and proposed nested cross-validation intervals. Bouthillier et al. [17] quantified the sources of variance (data split, initialisation, hyperparameter search) in machine learning benchmarks, and Varoquaux [18] showed how wide cross-validation error bars are in small samples. All these works assume independent observations.

Validation under spatial and temporal dependence.

In ecology, remote sensing and the geosciences, a large literature shows that random cross-validation is optimistic for spatially structured data and recommends spatial blocking, buffering or nearest-neighbour distance matching [5–7, 9, 10, 19, 20]. Wadoux et al. [21] argued that, when the goal is the accuracy of a map over a fixed region, design-based probability sampling is preferable to any form of spatial cross-validation. In time series, Bergmeir and Benítez [8] compared blocked and out-of-sample schemes and Bergmeir et al. [22] showed when standard cross-validation remains valid for autoregressive models. These works are concerned with the *point estimate* of predictive performance and its bias; the uncertainty of the comparison between two models, which is the object of this paper, is rarely addressed.

Comparing predictions of dependent data.

The Diebold–Mariano (DM) test [11], with the small-sample correction of Harvey et al. [23], compares the expected loss of two forecasts by applying a HAC (Newey–West [24]) variance estimator to the loss-differential series. Hering and Genton [12] extended it to spatial fields, estimating the covariance of the loss differential with a parametric variogram. HAC estimation under cross-sectional dependence goes back to Conley [25] and, for panels, to Driscoll and Kraay [26]; bandwidth choice and the poor size of HAC tests in small effective samples motivated the fixed-b theory of Kiefer and Vogelsang [27]. Block and stationary bootstraps [28–30] provide resampling alternatives. Two features distinguish our setting from that of the DM test: the predictions are produced by learners that are *refitted* on each training fold, which adds a training-induced variance component absent from forecast comparisons; and the dependence ranges of the loss differential are unknown and must be estimated without a parametric model of the data. DA-HAC addresses both.

Trustworthy machine learning and this special issue.

Recent contributions in this journal have advocated statistical tools for the robustness of machine learning models [31], practical safeguards against overfitting [32], and effect-size reporting beyond p-values in large data sets [33]. The present paper complements them by addressing the validity of the comparison step itself when observations are dependent.

3 Problem formulation

3.1 Setting and estimand

We observe n labelled examples $z_i = (x_i, y_i)$, $i = 1, \dots, n$, each attached to a spatial location $s_i \in \mathbb{R}^p$ (typically $p = 2$) and a time $t_i \in \mathbb{R}$; either component may be absent. Two learning procedures A and B (for instance, XGBoost and a random forest, or one learner with two feature sets) are to be compared with a loss ℓ, such as the squared error or the logarithmic loss [34].

The observations are grouped into G spatio-temporal blocks (for instance, square tiles crossed with time windows) and the blocks are allocated to K folds $F_1, \dots, F_K$, with $n_k = |F_k|$. For each fold k, both learners are fitted on a training set $T_k \subseteq \{i \notin F_k\}$, from which observations within a buffer distance of the fold may be removed, and are used to predict the observations of F_k. With $k(i)$ the fold of observation i, the out-of-fold *loss differential* is

$$d_i = \ell\big(y_i, \hat{f}_A^{(k(i))}(x_i)\big) - \ell\big(y_i, \hat{f}_B^{(k(i))}(x_i)\big), \tag{1}$$

so that negative values favour A, and the cross-validated estimate of the gain is $\hat{\Delta} = \bar{d} = n^{-1} \sum_i d_i$.

Definition 1 (Estimand) The *expected cross-validated loss differential* is $\Delta_n = \mathbb{E}[\hat{\Delta}]$, where the expectation is over the data-generating process and the randomness of the learners. It is the difference in expected loss between the two learning procedures, trained on about $n(1 - 1/K)$ observations, when predicting observations that are separated from the training data as prescribed by the blocking design.

The estimand depends on the validation design, as stressed by Bates et al. [16] for i.i.d. data and by the spatial validation literature [5, 21]: with random folds it describes interpolation, with spatial blocks it describes extrapolation to new areas. We take the design as given, chosen to match the intended use of the model, and focus on testing $H_0 : \Delta_n = 0$ against $H_1 : \Delta_n \neq 0$ and on confidence intervals for Δ_n.

3.2 A working model for loss differentials

The difficulty is that the d_i are dependent for two distinct reasons. We make them explicit with the following decomposition.

Definition 2 (Working model)

$$d_i = \Delta_n + \eta_{k(i)} + u_i, \tag{2}$$

where $(\eta_1, \dots, \eta_K)$ are exchangeable fold effects with $\mathrm{Var}(\eta_k) = \tau^2$ and $\mathrm{Corr}(\eta_k, \eta_l) = \rho$ for $k \neq l$, independent of u, and $u = (u_i)$ is a zero-mean, second-order stationary field with $\mathrm{Cov}(u_i, u_j) = C(s_i - s_j, t_i - t_j)$.

The fold effect η_k captures the variability induced by fitting the two learners on the training set T_k: an unlucky training sample, a different random seed or a different tree structure shifts the losses of all observations of the fold in the same direction. Because the training sets of different folds share a fraction $(K-2)/(K-1)$ of their observations, the fold effects are positively correlated. The field u captures test-sampling variability; it inherits the spatial and temporal autocorrelation of the residuals of both models, which is substantial whenever both models miss a spatially structured signal. Under (2),

$$\mathrm{Var}(\hat{\Delta}) = \underbrace{\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} C(s_i - s_j, t_i - t_j)}_{V_u} + \underbrace{\tau^2\Big[\sum_k \frac{n_k^2}{n^2} + \rho\sum_{k\neq l}\frac{n_k n_l}{n^2}\Big]}_{V_\eta}. \tag{3}$$

With equal fold sizes, $V_\eta = \tau^2[1/K + \rho(1-1/K)]$. Writing $V_u = \sigma_u^2/n_{\mathrm{eff}}$ defines the *effective sample size* $n_{\mathrm{eff}} = n/[1+(n-1)\bar{\rho}_u]$, where $\bar{\rho}_u$ is the average correlation between distinct u_i. Under moderate dependence n_{eff} can be orders of magnitude smaller than n.

3.3 Why conventional procedures fail

Procedures that assume independent observations.

The i.i.d. bootstrap of the d_i, the z-test on the d_i and McNemar-type tests estimate σ^2/n and therefore ignore both terms of (3) beyond the diagonal. Their variance is too small by the design-effect factor n/n_{eff}, and by V_η even without spatial dependence.

The paired t-test on fold scores.

The most common procedure computes the K fold means $m_k = n_k^{-1}\sum_{i\in F_k} d_i$ and refers $\bar{m}/(s_m/\sqrt{K})$ to a t_{K-1} distribution, with s_m^2 the sample variance of the m_k. Its weakness is captured by the following identity.

Proposition 1 *Let $m_1,\dots,m_K$ have common mean, average variance $\bar{v} = K^{-1}\sum_k \mathrm{Var}(m_k)$ and average pairwise covariance $\bar{c} = [K(K-1)]^{-1}\sum_{k\neq l}\mathrm{Cov}(m_k,m_l)$. Then*

$$\mathbb{E}\Big[\frac{s_m^2}{K}\Big] - \mathrm{Var}(\bar{m}) = -\bar{c}.$$

Proof $\mathrm{Var}(\bar{m}) = K^{-2}\big[\sum_k \mathrm{Var}(m_k) + \sum_{k\neq l}\mathrm{Cov}(m_k,m_l)\big] = [\bar{v} + (K-1)\bar{c}]/K$. Since the means are equal, $\mathbb{E}\sum_k (m_k-\bar{m})^2 = \sum_k \mathrm{Var}(m_k) - K\,\mathrm{Var}(\bar{m}) = (K-1)(\bar{v}-\bar{c})$, hence $\mathbb{E}[s_m^2]/K = (\bar{v}-\bar{c})/K$. Subtracting gives the result. □

The paired t-test is therefore valid only if fold means are uncorrelated. Under random K-fold cross-validation of dependent data, every fold is spread over the whole domain, the folds share the same spatial and temporal fluctuations, and $\bar{c}$ is close to $\mathrm{Var}(\bar{m})$ itself: the test then estimates only the small idiosyncratic part of the variance.

Overlapping training sets add $\rho\tau^2 > 0$ to $\bar{c}$, which is the mechanism identified by Dietterich [1] and corrected by Nadeau and Bengio [2] for i.i.d. data. Blocking reduces the spatial part of $\bar{c}$, but adjacent blocks of different folds remain correlated, and with $K = 5$ the test has only four degrees of freedom.

Cluster methods.

The block bootstrap and the random-intercept linear mixed model (LMM) treat the G blocks as independent clusters. They capture all within-block covariance, including the fold effects of (2), but ignore covariance between neighbouring blocks, which is non-negligible when the dependence range is comparable with the block size, and they ignore the correlation ρ of fold effects.

Time-only HAC.

Applying the Diebold–Mariano test with a Newey–West variance separately to the series of each site, as is often done for panels of forecasts, handles temporal dependence but ignores the correlation between sites at the same time and the fold effects.

4 Dependence-aware HAC inference

DA-HAC estimates the two components of (3) from the loss differentials of a single blocked cross-validation run. Figure 1 gives an overview of the complete procedure: blocked cross-validation produces the loss differentials (stage 1); two parallel branches estimate the test-sampling variance, through variogram-calibrated kernel weights, and the training-induced variance, from the dispersion of fold means (stage 2); the two are combined into a variance, a reference distribution and diagnostics (stage 3). The steps are summarised in Algorithm 1 and detailed below.

4.1 Fold-centred differentials

Let m_k be the mean of the d_i in fold k and $\tilde{u}_i = d_i - m_{k(i)}$. Under (2), $\tilde{u}_i = u_i - \bar{u}_{k(i)}$ is free of the fold effects, so that the dependence structure of u can be studied without confusion with the training-induced component.

4.2 Variogram-calibrated bandwidths

The key tuning parameter of any HAC estimator is the bandwidth, i.e. the distance beyond which covariances are neglected. Rules of thumb developed for time series [24, 35] do not transfer to space, and the dependence range of the loss differential is in general different from that of the response: it is the range of the *residual* structure that both models fail to capture. We therefore estimate it from $\tilde{u}$ directly. The directional empirical semivariograms

$$\hat{\gamma}_S(h) = \frac{1}{2|N_S(h)|} \sum_{(i,j)\in N_S(h)} (\tilde{u}_i - \tilde{u}_j)^2, \quad N_S(h) = \{(i,j) : t_i = t_j,\ \|s_i - s_j\| \in I_h\},$$

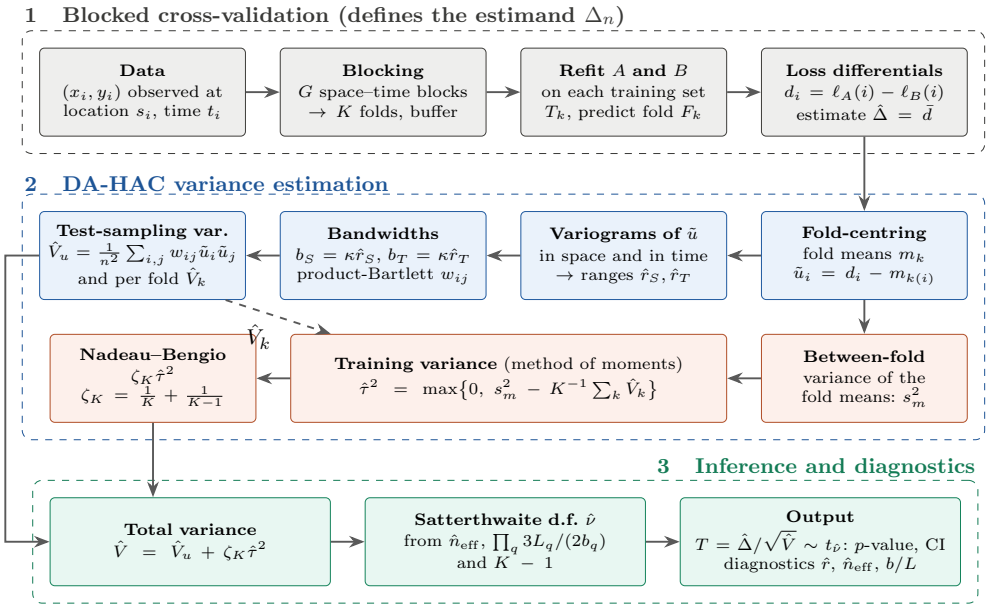


Fig. 1 Architecture of the proposed comparison procedure. Stage 1 (grey) is a standard blocked cross-validation that yields the per-observation loss differentials d_i and defines the estimand. Stage 2 (DA-HAC) estimates the test-sampling variance $\hat{V}_u$ (blue branch, read from right to left: fold-centring, variograms, bandwidths, product-Bartlett kernel) and the training-induced variance $\zeta_K\hat{\tau}^2$ (orange branch). Stage 3 (green) combines them into the total variance, the Satterthwaite degrees of freedom and the test, together with the diagnostics used in Section 8.

$$\hat{\gamma}_T(\ell) = \frac{1}{2|N_T(\ell)|}\sum_{(i,j)\in N_T(\ell)} (\tilde{u}_i - \tilde{u}_j)^2, \qquad N_T(\ell) = \{(i,j) : s_i = s_j,\ |t_i - t_j| \in I_\ell\},$$

are computed over distance and lag bins I_h, I_ℓ up to a maximum lag (by default half the spatial extent and a quarter of the temporal extent). For purely spatial data, $N_S(h)$ uses all pairs. To each we fit the exponential model $\gamma(h) = c_0 + c_1\{1 - \exp(-h/a)\}$ by the weighted least squares of Cressie [36], with weights $|N(h)|/\gamma(h)^2$. The *practical range* $r = 3a$ is the distance at which the correlation of the structured component falls to 5%; it is set to zero when the structured part explains less than 5% of the sill ($c_1 < 0.05(c_0 + c_1)$). The spatial and temporal bandwidths are

$$b_S = \kappa\,\hat{r}_S, \qquad b_T = \kappa\,\hat{r}_T,$$

with $\kappa = 1$ chosen in a pilot study on separate random seeds (Section 5). We also report the *range ratio* $\max_q b_q/L_q$, the largest bandwidth relative to the extent of the domain along the same axis, as a diagnostic: when it approaches one, the sample contains only a handful of effectively independent regions, and no procedure can estimate the variance of $\hat{\Delta}$ reliably from it.

4.3 Kernel estimate of the test-sampling variance

We use the product-Bartlett kernel

$$w_{ij} = \prod_{q=1}^{p} \Big(1 - \frac{|s_{iq} - s_{jq}|}{b_S}\Big)_+ \Big(1 - \frac{|t_i - t_j|}{b_T}\Big)_+, \tag{4}$$

where a zero bandwidth keeps only pairs with identical coordinates on that axis. Each factor is a one-dimensional Bartlett (triangular) kernel, whose Fourier transform is non-negative; the product is therefore a positive semi-definite kernel on $\mathbb{R}^{p+1}$, which guarantees non-negative variance estimates. (A Bartlett kernel applied to the Euclidean distance in two or more dimensions does not have this property.) The test-sampling variance and its within-fold analogues are

$$\hat{V}_u = \frac{1}{n^2} \sum_{i,j} w_{ij} \tilde{u}_i \tilde{u}_j, \qquad \hat{V}_k = \frac{1}{n_k^2} \sum_{i,j \in F_k} w_{ij} \tilde{u}_i \tilde{u}_j. \tag{5}$$

Only pairs with $w_{ij} > 0$ contribute; they are enumerated with a k-d tree in the Chebyshev metric after scaling each coordinate by its bandwidth, at a cost proportional to the number of such pairs.

4.4 Training-induced variance

Under (2), the between-fold variance of fold means exceeds their within-fold test-sampling variance by the variance of the fold effects. We estimate it by the method of moments,

$$\hat{\tau}^2 = \max\Big\{0,\ s_m^2 - \frac{1}{K} \sum_{k=1}^{K} \hat{V}_k \Big\}. \tag{6}$$

Since $\mathbb{E}[s_m^2]$ subtracts the covariance between fold effects (Proposition 1), $\hat{\tau}^2$ targets $\tau^2(1-\rho)$ rather than τ^2. The correlation ρ is not identifiable from one cross-validation run [13]; following Nadeau and Bengio [2] we set it to the test fraction $\rho = n_{\text{test}}/n = 1/K$. Then $V_\eta = \tau^2[1/K + \rho(1 - 1/K)] = \zeta_K \, \tau^2(1 - \rho)$ with

$$\zeta_K = \frac{1}{K} + \frac{1}{K-1},$$

which is exactly the Nadeau–Bengio inflation $1/J + n_{\text{test}}/n_{\text{train}}$ with $J = K$ splits. The final variance estimate is

$$\hat{V} = \hat{V}_u + \zeta_K \hat{\tau}^2. \tag{7}$$

Fold-centring removes from $\tilde{u}$ the fold-level average of u itself, so $\hat{V}_u$ is slightly biased downward when the dependence range is comparable to the fold size; because s_m^2 contains that same component while the $\hat{V}_k$ do not, the deficit is transferred to $\hat{\tau}^2$, which partially compensates for it.

Algorithm 1 DA-HAC test for $H_0 : \Delta_n = 0$

Require: loss differentials d_i, folds $k(i)$, coordinates (s_i, t_i), level α, multiplier κ (default 1)

1: $m_k \leftarrow$ mean of d_i over F_k; $\tilde{u}_i \leftarrow d_i - m_{k(i)}$; $\hat{\Delta} \leftarrow n^{-1} \sum_i d_i$
2: Fit exponential models to the spatial and temporal empirical variograms of $\tilde{u}$; $b_S \leftarrow \kappa \hat{r}_S$, $b_T \leftarrow \kappa \hat{r}_T$
3: Enumerate pairs with $w_{ij} > 0$ under the product-Bartlett kernel (4)
4: Compute $\hat{V}_u$ and $\hat{V}_1, \ldots, \hat{V}_K$ by (5)
5: $\hat{\tau}^2 \leftarrow \max\{0, s_m^2 - K^{-1} \sum_k \hat{V}_k\}$; $\hat{V} \leftarrow \hat{V}_u + (1/K + 1/(K-1))\hat{\tau}^2$
6: Compute $\hat{n}_{\text{eff}}$, ν_u and $\hat{\nu}$ by (8), and the range ratio $\max_q b_q / L_q$
7: **return** $p = 2 \Pr(t_{\hat{\nu}} > |\hat{\Delta}|/\sqrt{\hat{V}})$ and the interval $\hat{\Delta} \pm t_{\hat{\nu}, 1-\alpha/2} \sqrt{\hat{V}}$

4.5 Reference distribution

HAC tests are known to over-reject when the effective sample size is small [27], and $\hat{\tau}^2$ rests on only $K - 1$ degrees of freedom. We approximate the distribution of $\hat{V}$ by a scaled χ^2 with Welch–Satterthwaite [37] degrees of freedom

$$\hat{\nu} = \frac{\hat{V}^2}{\hat{V}_u^2/\nu_u + (\zeta_K \hat{\tau}^2)^2/(K-1)}, \qquad \nu_u = \min\Big\{\hat{n}_{\text{eff}} - 1,\ \prod_q \frac{3L_q}{2b_q}\Big\}, \tag{8}$$

with $\hat{n}_{\text{eff}} = n^{-1} \sum_i \tilde{u}_i^2 / \hat{V}_u$ truncated to $[2, n]$, where the product runs over the spatial and temporal axes with a positive bandwidth, and L_q is the extent of the domain along axis q. The second term of ν_u is the classical approximation to the degrees of freedom of a kernel spectral estimate [35]: for a lag window k of bandwidth b on a record of length L, $\mathrm{Var}(\hat{V})/V^2 \approx 2(b/L) \int k^2$, with $\int k^2 = 2/3$ for the Bartlett window, and the factors multiply for a product kernel. It caps the degrees of freedom when the bandwidth is not small compared with the domain, the regime in which HAC tests are most anti-conservative. The test statistic $T = \hat{\Delta}/\sqrt{\hat{V}}$ is referred to a Student $t_{\hat{\nu}}$ distribution, and the $(1 - \alpha)$ confidence interval is $\hat{\Delta} \pm t_{\hat{\nu}, 1-\alpha/2} \sqrt{\hat{V}}$.

4.6 Properties

Remark 1 (Special cases) If $b_S = b_T = 0$ and $\hat{\tau}^2 = 0$, then $\hat{V} = n^{-2} \sum_i \tilde{u}_i^2$ and DA-HAC reduces to the one-sample t-test on the d_i. If the data have no spatial (or no temporal) coordinate, the corresponding factor of (4) is omitted. Because the kernel is positive semi-definite, $\hat{V} \geq 0$ always.

Remark 2 (Consistency) For fixed bandwidths, $\hat{V}_u$ is a Conley-type spatial HAC estimator [25]. Under increasing-domain asymptotics with mixing conditions and bandwidths that grow with, but more slowly than, the size of the domain, such estimators are consistent [25, 35]. Our bandwidths are data-dependent and our variance includes a component, $\hat{\tau}^2$, estimated from only K folds; we therefore do not claim exact asymptotic validity, and instead evaluate the finite-sample behaviour of the whole procedure by simulation with known truth.

Remark 3 (Scope) DA-HAC requires only the per-observation losses of the two models and their coordinates; it is agnostic to the learners, to the loss and to the blocking scheme. The same construction applies to any additive performance measure (squared error, absolute error, log loss, Brier score, 0–1 loss). Non-additive measures such as the AUC or the F1-score would require a linearisation (influence function) first.

5 Simulation study I: controlled loss-differential fields

The first study isolates the inferential problem from the learning problem. Prediction errors are simulated directly, so that the true expected loss differential is known exactly and the dependence of the d_i is controlled.

5.1 Design

Fields.

Observations lie on a 12×12 grid of sites observed at 24 time points ($n = 3456$). All fields are separable Gaussian fields with unit variance, exponential spatial correlation $\exp(-h/a)$ and AR(1) temporal dependence with coefficient ϕ. Four dependence levels are considered: *none* ($a = 0$, $\phi = 0$), *weak* ($a = 1$, $\phi = 0.3$), *moderate* ($a = 2$, $\phi = 0.6$) and *strong* ($a = 4$, $\phi = 0.8$). The corresponding practical ranges are $3a = 0, 3, 6, 12$ grid units in space and $-3/\log\phi = 0, 2.5, 5.9, 13.4$ time steps; at the strong level the range is as large as the domain.

Losses.

With ε, e_A and e_B three independent fields, the response noise is ε, model A has errors e_A and model B has errors $\sqrt{1+\delta}\, e_B$. Under squared loss,

$$d_i = (\varepsilon_i - e_{A,i})^2 - (\varepsilon_i - \sqrt{1+\delta}\, e_{B,i})^2 + \eta_{k(i)}, \qquad \mathbb{E}[d_i] = -\delta,$$

so that $\Delta_n = -\delta$ exactly. The fold effects $\eta_k = \sigma_\eta(\sqrt{\rho}\, g + \sqrt{1-\rho}\, h_k)$, with $g, h_1, \ldots, h_K$ i.i.d. standard normal, $\sigma_\eta = 0.05$ and $\rho = 0.25$, mimic the training-induced variability of (2), including its correlation across folds. We use $\delta = 0$ (null) and $\delta \in \{0.1, 0.2, 0.4\}$ (alternatives).

Validation and procedures.

The domain is cut into 4×4-site tiles crossed with 6-step time windows ($G = 36$ blocks), which are allocated to $K = 5$ folds with balanced sizes. We compare the paired t-test on fold means, the i.i.d. bootstrap, the block bootstrap and a random-intercept LMM (both with the 36 blocks as clusters), the Diebold–Mariano test with a site-wise Newey–West variance (DM–HAC), and DA-HAC. Three ablations of DA-HAC are included: without the training term ($\hat{\tau}^2 = 0$), with a normal reference distribution, and with a fixed bandwidth equal to the block size instead of the variogram-calibrated one. Bootstraps use 999 resamples. There are 1000 replicates per dependence level under the null and 300 under each alternative, so that the Monte Carlo standard error of a rejection rate of 5% is 0.7 percentage points.

Pilot.

The bandwidth multiplier κ and the reference distribution were chosen in a pilot study with 300 null replicates per level, run on random seeds disjoint from those of the main study. Rejection rates of DA-HAC varied by at most 2.7 percentage points across $\kappa \in \{0.5, 1, 1.5, 2\}$ at any level, with no systematic trend, and we retained $\kappa = 1$. Capping the degrees of freedom by the kernel term of (8) reduced the rejection rate at the strong level from 17.7% to 15.3% without affecting the other levels by more than 1.4 points, and was retained. The pilot results are included in the code repository.

5.2 Results

Type I error.

Table 1 and Fig. 2 report rejection rates under the null. Procedures that ignore dependence fail quickly: the i.i.d. bootstrap rejects a true null in 19% of replicates under weak dependence and in 80% under strong dependence, and the time-only DM–HAC test, which ignores correlation between sites, behaves almost as badly (17% and 70%). The two cluster procedures, block bootstrap and LMM, are much better but still drift upwards as the range of dependence approaches the block size (15% and 14% at the moderate level, 27% and 25% at the strong level), because they treat neighbouring blocks as independent. Even without any spatial or temporal dependence they reject 9% and 7% of true nulls, as they do not account for the correlation of fold effects. The paired t-test on blocked fold means is the best conventional procedure, in line with Proposition 1, since blocking removes most of the covariance between fold means; it nevertheless reaches 11% at the moderate and 18% at the strong level.

DA-HAC has rejection rates of 4.8%, 7.2%, 9.9% and 12.8% at the four levels. It is the procedure closest to the nominal level without dependence and under moderate and strong dependence; under weak dependence the paired t-test (6.7%) is marginally closer. Its ablation with a fixed bandwidth equal to the block size is slightly better at the weak and moderate levels (5.6% and 8.8%). That variant uses knowledge of a sensible block size, which the variogram calibration recovers from the data. DA-HAC is *not* exactly calibrated under moderate and strong dependence. At the strong level the practical range (12 sites and 13 time steps) equals the extent of the domain; the median estimated range ratio b/L is 0.52, the sample contains only a few effectively independent regions, and the estimated ranges are about half the true ones because the variogram cannot be estimated beyond the domain. In this regime no method we studied is reliable, and the range ratio is an informative warning (Section 8).

Ablations.

Removing the training term $\zeta_K \hat{\tau}^2$ almost triples the rejection rate at the weak and moderate levels (14% and 24% instead of 7% and 10%): fold-centred kernel estimators miss the between-fold component of the variance, whose estimation is therefore essential. Replacing the Satterthwaite reference by a normal distribution adds up to 3 percentage points, the largest increase being at the strong level.

Table 1 Experiment 1: rejection rate (%) of a true null at $\alpha = 0.05$ (1000 replicates per level) and coverage (%) of the 95% confidence interval (all 1900 replicates per level). Bold: rejection rate outside the 95% Monte Carlo band $[3.6, 6.4]$ around 5%.

	Type I error (%)				Coverage (%)			
Procedure	None	Weak	Mod.	Strong	None	Weak	Mod.	Strong
Paired t (folds)	**6.6**	**6.7**	**10.7**	**17.9**	93.5	93.6	89.2	81.4
i.i.d. bootstrap	**8.1**	**19.4**	**57.4**	**80.3**	91.8	79.7	43.0	19.3
Block bootstrap	**9.2**	**8.7**	**15.3**	**27.2**	91.1	90.5	82.6	70.4
Mixed model	**7.4**	**8.0**	**13.6**	**24.5**	92.3	91.7	85.7	74.2
DM–HAC (time only)	**8.4**	**17.0**	**48.2**	**70.2**	91.8	82.4	51.8	28.1
DA-HAC (proposed)	4.8	**7.2**	**9.9**	**12.8**	95.4	92.6	89.7	85.6
DA-HAC without $\hat{\tau}^2$ term	**7.1**	**14.1**	**23.6**	**25.6**	93.1	85.3	75.8	73.2
DA-HAC, normal reference	5.8	**7.8**	**10.8**	**15.6**	94.4	92.1	88.7	82.8
HAC, block-size bandwidth	6.3	5.6	**8.8**	**14.3**	94.1	94.3	90.5	83.6

Table 2 Experiment 1: size-adjusted power (%) for true loss gaps $\delta \in \{0.1, 0.2, 0.4\}$ (300 replicates each). The critical value of each procedure is the 5% quantile of its own null p-values; it is undefined (–) when more than 5% of null bootstrap p-values equal the smallest attainable value.

	No dependence			Moderate			Strong		
Procedure	0.1	0.2	0.4	0.1	0.2	0.4	0.1	0.2	0.4
Paired t (folds)	24	54	96	7	13	29	5	7	9
i.i.d. bootstrap	31	82	100	–	–	–	–	–	–
Block bootstrap	29	81	100	9	18	46	9	12	17
Mixed model	30	81	100	10	20	49	5	8	12
DM–HAC (time only)	31	83	100	10	16	46	7	8	10
DA-HAC (proposed)	26	61	89	7	14	29	5	7	12

Coverage, width and power.

Coverage of the 95% intervals over all replicates (null and alternatives, Table 1) mirrors the type I error: 95.4%, 92.6%, 89.7% and 85.6% for DA-HAC, against 79.7%, 43.0% and 19.3% for the i.i.d. bootstrap from the weak level on. The DA-HAC intervals are wider than those of the cluster procedures (Fig. 2b), which is the price of validity: the narrow intervals of the i.i.d. bootstrap and DM–HAC do not widen with dependence at all. Raw rejection rates under the alternatives (Fig. 3) are not comparable across procedures whose sizes differ by an order of magnitude; we therefore also report size-adjusted power, using as critical value the 5% quantile of each procedure's null p-values (Table 2). DA-HAC has about the same size-adjusted power as the paired t-test. Without dependence it is less powerful than the cluster procedures (61% against 81% at $\delta = 0.2$), because the training term is estimated from only $K - 1 = 4$ degrees of freedom; under moderate dependence the cluster procedures retain more size-adjusted power, but this advantage cannot be realised in practice, since their nominal tests over-reject and their null distribution is unknown.

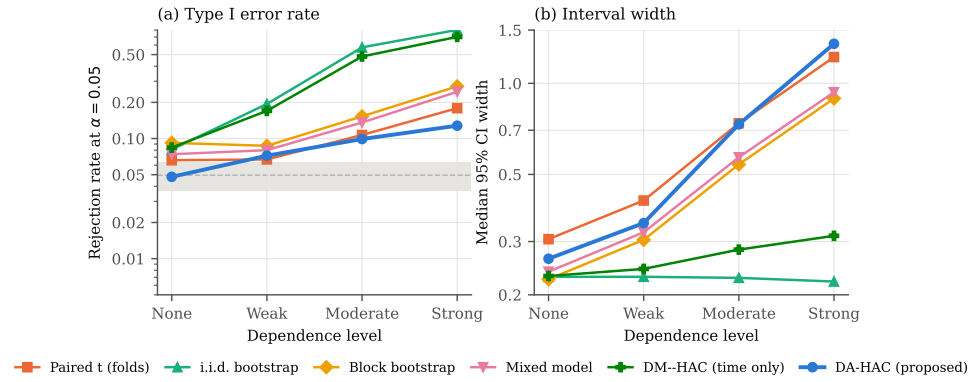


Fig. 2 Experiment 1. (a) Rejection rate of a true null hypothesis (log scale); the dashed line is the nominal level and the grey band the 95% Monte Carlo band. (b) Median width of the 95% confidence interval under the null.

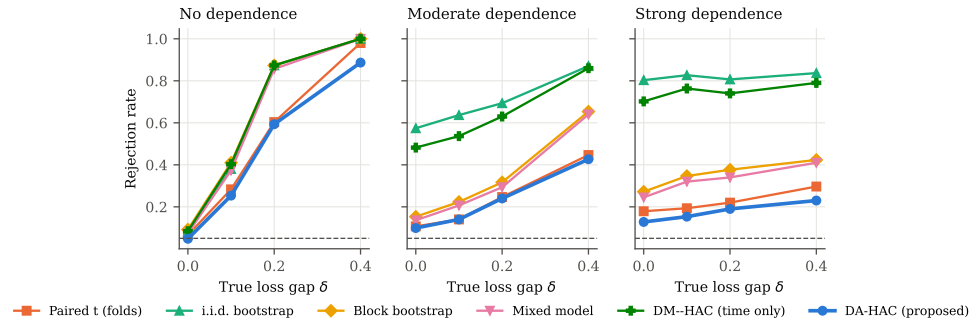


Fig. 3 Experiment 1. Raw rejection rate as a function of the true loss gap δ ($\delta = 0$ is the null). High curves at $\delta = 0$ indicate invalid tests, not power.

6 Simulation study II: learners on spatio-temporal data

The second study reproduces the full pipeline of an applied comparison: data generation, model fitting by cross-validation and testing. The difficulty in such a study is to know the truth. We obtain an exact null by exchangeability.

6.1 Design

Data.

On a 10×10 grid observed at 20 time points ($n = 2000$), three covariates x_1, x_2, x_3 and a latent residual w are independent spatio-temporal fields with the dependence

levels of Section 5. The response is

$$y = g(x_1) + (1-\gamma)\, g(x_2) + x_3 + w + \epsilon, \qquad g(v) = \sin(2v) + v, \qquad \epsilon \sim N(0, 0.5^2) \text{ i.i.d.}$$

Models.

Model A is XGBoost [38] (100 trees of depth 4, learning rate 0.1, row subsampling 0.8) fitted on (x_1, x_3, s, t); model B is the same learner fitted on (x_2, x_3, s, t). Including the coordinates lets both models exploit the spatial and temporal structure of w, as is common in practice.

Exact null by exchangeability.

When $\gamma = 0$, the joint distribution of the data is invariant under the exchange of x_1 and x_2, which exchanges the roles of A and B. Hence $\mathbb{E}[d_i] = 0$ for every observation and every validation design: the two models have exactly the same expected performance, although in any given sample one of them will appear better. When $\gamma > 0$, x_1 carries more signal than x_2 and A is truly better; we use $\gamma \in \{0.1, 0.2, 0.3\}$.

Validation and procedures.

Two pipelines are applied to each data set. The *conventional* pipeline uses random 5-fold cross-validation with the paired t-test on fold means and the i.i.d. bootstrap, and 5×5 repeated random cross-validation with the corrected resampled t-test [2]. The *dependence-aware* pipeline uses blocked 5-fold cross-validation (5×5-site tiles crossed with 4-step windows, $G = 20$ blocks, and a buffer that removes from the training set observations within one site and one time step of the test fold) with the procedures of Section 5. There are 300 replicates per dependence level under the null and 150 under each alternative.

6.2 Results

Random cross-validation.

The left panel of Fig. 4 and the upper part of Table 3 show that the conventional pipeline fails as soon as the data are dependent. With random 5-fold cross-validation, the paired t-test rejects a true null in 15% of replicates under moderate and 29% under strong dependence, and the i.i.d. bootstrap in 24% and 40%. The corrected resampled t-test of Nadeau and Bengio is well calibrated without dependence (3.7%) and under weak dependence (5.7%), as designed, but not beyond (16% and 33%): its correction accounts for overlapping training sets, not for dependent test observations. Note also that even without spatio-temporal dependence the paired t-test rejects 9% of true nulls, the problem identified by Dietterich [1].

The estimand changes with the validation design.

Random cross-validation does not only produce invalid tests; it also estimates a different quantity. Table 4 reports the average estimated gain of model A under both designs. Because both models can exploit the coordinates to interpolate between nearby training observations, random cross-validation attenuates the advantage conferred by the

Table 3 Experiment 2: rejection rate (%) of a true null ($\gamma = 0$, 300 replicates per level) and raw power (%) at the moderate dependence level (150 replicates per value of γ). Bold: rejection rate outside the 95% Monte Carlo band $[2.5, 7.5]$.

		Type I error (%)				Power, moderate (%)		
CV	Procedure	None	Weak	Mod.	Strong	$\gamma = 0.1$	0.2	0.3
Random	Paired t (folds)	**9.0**	**8.3**	**14.7**	**29.3**	67.3	99.3	100.0
Random	i.i.d. bootstrap	7.0	**12.7**	**23.7**	**40.0**	80.0	99.3	100.0
Random	Corrected resampled t	3.7	5.7	**16.3**	**32.7**	71.3	100.0	100.0
Blocked	Paired t (folds)	5.3	**7.7**	**14.3**	**14.3**	24.7	66.7	89.3
Blocked	i.i.d. bootstrap	5.3	**22.7**	**55.0**	**66.3**	77.3	96.0	99.3
Blocked	Block bootstrap	**10.7**	**12.3**	**20.3**	**28.0**	42.7	80.7	98.0
Blocked	Mixed model	**8.0**	**10.3**	**18.0**	**21.3**	38.0	76.0	97.3
Blocked	DM–HAC (time only)	6.3	**20.7**	**49.7**	**59.3**	72.0	93.3	99.3
Blocked	DA-HAC (proposed)	3.3	7.0	**15.0**	**14.3**	23.3	62.0	80.0

more informative covariate, by about 30% under moderate and 50% under strong dependence (-0.53 instead of -0.80, and -0.30 instead of -0.60, for $\gamma = 0.2$). A gain measured by random cross-validation is thus not a conservative estimate of the gain at new locations and times: it is an estimate of a different, interpolation-based, quantity.

Blocked cross-validation.

Blocking is necessary but not sufficient. With blocked folds, procedures that assume independent observations (i.i.d. bootstrap, 55% at the moderate level) or ignore cross-sectional correlation (DM–HAC, 50%) are even worse than under random cross-validation, because the dependent test errors are no longer diluted across folds. The cluster procedures over-reject at every level, including without dependence (11% and 8%), since the fold effects induced by refitting XGBoost are correlated across blocks of the same fold. DA-HAC and the paired t-test on blocked folds are the two best procedures: they are calibrated without dependence (3.3% and 5.3%) and nearly so under weak dependence (7.0% and 7.7%), but both reject about 15% of true nulls under moderate and strong dependence. In this design the dependence of the loss differential has an additional, domain-wide component: whichever of the two covariate fields happens to be more variable over the whole domain in a given sample favours the corresponding model in every fold. With a practical range of a quarter to a half of the domain, this component is shared by all blocks and is only weakly identifiable from a single sample. We return to this limitation in Section 8.

Power.

Under the alternatives (right part of Table 3), the blocked paired t-test and DA-HAC have similar power (67% and 62% for $\gamma = 0.2$ at the moderate level), lower than that of the invalid procedures. Without dependence, DA-HAC is less powerful than the paired t-test for large effects (91% against 100% for $\gamma = 0.3$), because a large fold-level variance $\hat{\tau}^2$, estimated with $K - 1$ degrees of freedom, then dominates its variance and reduces its degrees of freedom.

Table 4 Experiment 2: Monte Carlo mean of the estimated loss differential $\hat{\Delta}$ (model A minus model B, squared loss) under random (R) and blocked (B) cross-validation. Under the null ($\gamma = 0$) both are zero by exchangeability.

	$\gamma = 0$		$\gamma = 0.1$		$\gamma = 0.2$		$\gamma = 0.3$	
Dependence	R	B	R	B	R	B	R	B
None	-0.003	0.002	-0.395	-0.401	-0.809	-0.827	-1.114	-1.133
Weak	-0.012	-0.013	-0.356	-0.425	-0.699	-0.821	-1.022	-1.205
Moderate	-0.009	-0.031	-0.275	-0.403	-0.528	-0.797	-0.747	-1.145
Strong	0.000	0.020	-0.155	-0.293	-0.297	-0.597	-0.424	-0.847

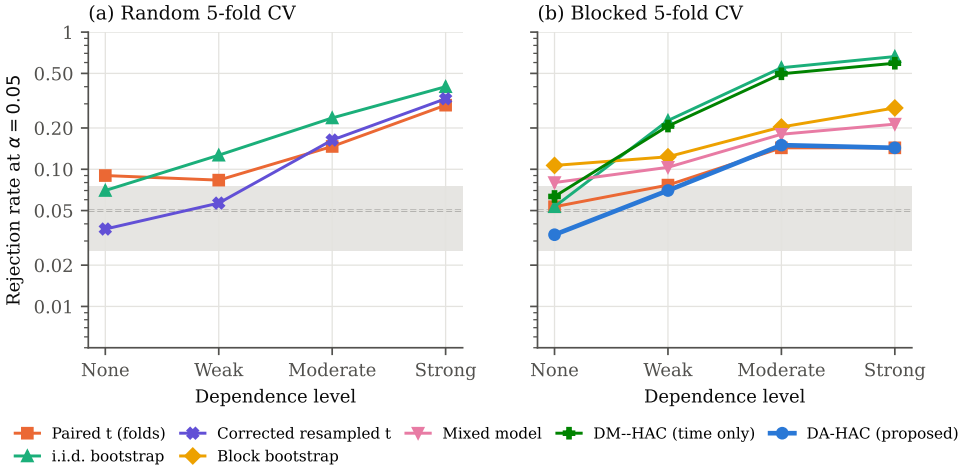


Fig. 4 Experiment 2. Rejection rate of a true null hypothesis (two XGBoost models with exchangeable feature sets) under random and blocked cross-validation (log scale). Dashed line: nominal level; grey band: 95% Monte Carlo band.

7 Case studies

7.1 Data and protocol

We use three public data sets obtained from the Rdatasets collection [39].

NYC weather.

Hourly weather records at the three New York City airports (EWR, JFK, LGA) in 2013, distributed with the `nycflights13` package [40] ($n = 26{,}110$ after removing five records with missing temperature, humidity or wind speed). The task is to classify whether precipitation was recorded in the hour, from temperature, dew point, humidity, wind direction and speed, pressure, month, hour and airport; 6.7% of hours are positive. The loss is the logarithmic loss.

Melbourne pedestrians.

Hourly pedestrian counts at four sensors in Melbourne in 2015–2016, distributed with the `tsibble` package [41] (n = 66,037). The task is to predict $\log(1 + \text{count})$ from hour, day of week, month, day of year, year and sensor, under squared loss.

Las Rosas maize.

Yield-monitor data from a maize field in Las Rosas, Argentina, in 1999 and 2001 [42], distributed with the `agridat` package ($n = 3443$, field of about 660×380 m). The task is to predict yield from the nitrogen rate, brightness value, topographic zone, year and coordinates, under squared loss.

For the two hourly data sets, blocks are two-week windows (26 and 53 blocks); as the airports lie within about 35 km of one another and the sensors within about 2 km, they are treated as co-located, so that all sites observed at the same time may be correlated, and a 24-hour buffer is removed from the training set on each side of a test block. For the maize data, blocks are 75×75 m tiles (35 non-empty blocks) with a 10 m buffer. We compare XGBoost (300 trees of depth 6, learning rate 0.05, row and column subsampling 0.8) with a random forest [43] (200 trees, minimum leaf size 5, 50% of features per split) and with a linear baseline (logistic or ridge regression on standardised, mean-imputed features). All models use scikit-learn [44] or the XGBoost library with default settings otherwise; no hyperparameter was tuned, since our object is the comparison, not the models.

Seed-null diagnostic.

For each data set we also compare XGBoost with itself, the two runs differing only in their random seed (30 pairs). By symmetry, the expected loss differential of such a comparison is zero, while in each run the stochastic subsampling produces a non-zero observed difference. A procedure that frequently declares one seed "significantly better" than the other confuses training noise with a real improvement. Because the data are held fixed, this diagnostic measures the sensitivity of each procedure to training randomness only; it is not a full assessment of size, which Sections 5 and 6 provide.

7.2 Results

Table 5 and Fig. 5 report the estimated loss differentials (XGBoost minus competitor; negative values favour XGBoost), their 95% intervals and the p-values; Table 6 gives the DA-HAC diagnostics and Fig. 6 the variograms of the fold-centred loss differentials.

NYC weather: an improvement that vanishes.

Under random cross-validation, XGBoost appears clearly better than the random forest (log-loss gain 0.011, $p \leq 0.001$ with every procedure) and than logistic regression (gain 0.041, $p \leq 0.001$). Under two-week blocked cross-validation both gains disappear: the point estimates become slightly *positive* (+0.0036 and +0.0028), and DA-HAC, the paired t-test and the cluster procedures find no significant difference ($p \geq 0.28$). Random folds let the models recognise weather episodes from neighbouring hours of the same day; the apparent superiority of XGBoost reflects its better use of this

leakage, not better prediction of new periods. The i.i.d. bootstrap applied to the blocked differentials even declares the random forest significantly *better* than XGBoost ($p = 0.023$), a reversal produced by ignoring the 20-hour dependence range of the loss differential (Table 6).

Melbourne pedestrians: a small gain of uncertain reality.

XGBoost and the random forest are indistinguishable under random cross-validation, but under blocked cross-validation XGBoost has a small advantage (0.0025 in squared error of the log count). The paired t-test ($p = 0.033$) and the i.i.d. bootstrap ($p = 0.011$) call it significant; DA-HAC ($p = 0.17$), DM–HAC ($p = 0.16$) and the cluster procedures ($p \geq 0.40$) do not. The variogram of the loss differential has a short estimated range (5 hours) but pedestrian traffic also has daily and weekly cycles and holiday effects, which an exponential variogram does not represent; the wider block-bootstrap interval suggests that some of this longer-range dependence is missed by the kernel estimator. We would conclude that the gain is not established. By contrast, the advantage of XGBoost over the linear baseline (1.85) is overwhelming under every procedure.

Las Rosas maize: when the diagnostics warn.

The gain of XGBoost over the random forest is $2.8\,(\text{q/ha})^2$ under random cross-validation, significant with all procedures ($p \leq 0.013$), and much larger but much more uncertain under spatial blocking (21). The i.i.d. bootstrap ($p = 0.001$) and the block bootstrap ($p = 0.043$) declare it significant, the LMM is borderline ($p = 0.069$), and the paired t-test ($p = 0.31$) and DA-HAC ($p = 0.45$) do not. The DA-HAC diagnostics explain why: the loss differential has a spatial range of about 150 m in a field of 660×380 m (range ratio 0.40), so the effective sample size is about 150 for 3443 yield-monitor points, and the training term dominates the variance ($\hat{\nu} = 4.7$). Following our recommendations, this comparison is unresolved: the data contain too few independent regions to tell whether gradient boosting is genuinely better than the random forest on this field. The gain over the linear model (160, interval $[110, 210]$) is, again, unambiguous.

Seed-null diagnostic.

Table 7 reports how often each procedure declares one of two XGBoost runs, differing only by their seed, significantly better than the other. Procedures that resample observations independently do so in 33–87% of the pairs under blocked cross-validation, and in 43% of the pairs for the pedestrian data under random cross-validation, where the subsampling noise of the learner is spatially and temporally coherent. DM–HAC does so in 23–33%. DA-HAC (0–10%) and the cluster procedures (0–10%) are the least sensitive to training noise, followed by the blocked paired t-test (0–13%). With 30 pairs per data set, these rates have a Monte Carlo standard error of up to 9 percentage points and should be read qualitatively.

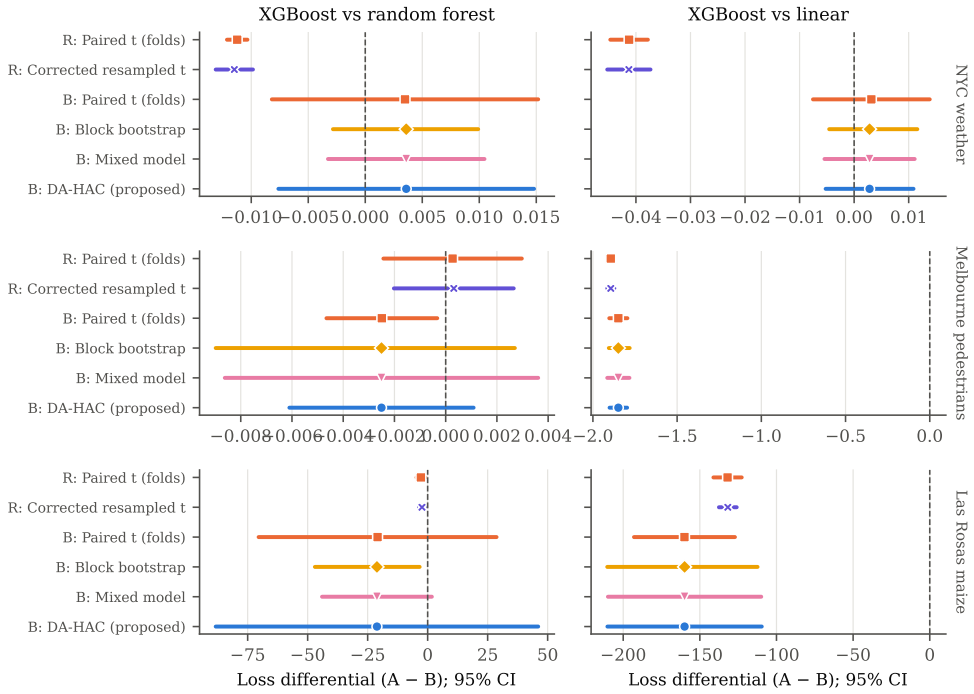


Fig. 5 Case studies: 95% confidence intervals for the loss differential (XGBoost minus competitor) under random (R) and blocked (B) cross-validation. The dashed line marks equal performance.

8 Discussion

8.1 Are machine learning improvements statistically real?

Our results give a nuanced answer to the question in the title. With spatially or temporally dependent data, the conventional pipeline (random K-fold cross-validation followed by a paired t-test, an i.i.d. bootstrap, or even the corrected resampled t-test) produces false discoveries at rates between 15% and 40% under moderate to strong dependence in our simulations, instead of the nominal 5%. Moreover, random cross-validation targets an interpolation estimand that can understate the gain at new places and times by half. Many reported "significant" improvements on such data are therefore not statistically supported, and some genuine improvements are misestimated.

Switching to blocked cross-validation fixes the estimand but not, by itself, the inference: applied to blocked folds, procedures that resample observations or that model only temporal dependence are *more* anti-conservative than under random folds (60–80% false positives under strong dependence). Cluster procedures (block bootstrap, random-intercept models) are a clear improvement but remain anti-conservative whenever the dependence range is not small compared with the blocks, and they ignore the correlation of training-induced fold effects. The two procedures that best control

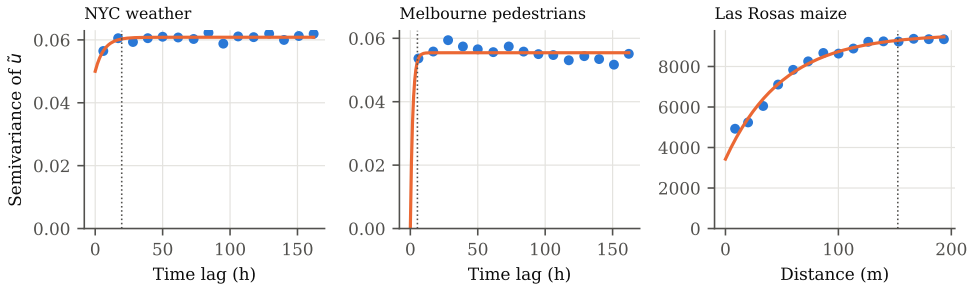


Fig. 6 Case studies: empirical semivariograms of the fold-centred loss differentials $\tilde{u}$ (XGBoost vs random forest, blocked CV; points), fitted exponential models (lines) and practical ranges (dotted).

the error rate are the simple paired t-test on blocked fold means, whose validity is explained by Proposition 1, and DA-HAC.

8.2 What DA-HAC adds, and what it does not

DA-HAC was the procedure closest to the nominal level in Experiment 1 at three of the four dependence levels, and tied with the blocked paired t-test in Experiment 2. Its additional value over the paired t-test lies less in raw calibration than in four practical properties. (i) It does not depend on the number of folds: the paired t-test with $K = 5$ has four degrees of freedom whatever the sample size, and its validity hinges on the absence of correlation between fold means (Proposition 1), which blocking only approximates. (ii) It decomposes the uncertainty into a test-sampling component $\hat{V}_u$ and a training component $\zeta_K \hat{\tau}^2$, which tells the analyst whether more data or more training runs would sharpen the comparison. (iii) It reports interpretable diagnostics: estimated dependence ranges, effective sample size and range ratio. (iv) The ablations show that its two ingredients are both needed: without the training term its error rate is up to 2.4 times larger, and a fixed bandwidth requires knowing a sensible dependence range in advance.

DA-HAC does not solve the problem completely. When the dependence range is a sizeable fraction of the domain (median range ratio above about 0.25 in our experiments), DA-HAC rejected a true null in 10–15% of cases, the blocked paired t-test in 11–18%, and the other procedures more often still. This is not a defect of a particular estimator but a limit of information: a sample that contains only a handful of effectively independent regions cannot tell whether a difference between two models is systematic or due to the particular realisation of the large-scale fields in that region. Fixed-b asymptotics [27] and our kernel-based degrees of freedom mitigate, but do not remove, this problem. In such cases the honest conclusion is that the data do not allow a reliable comparison, and the remedy is more data over a larger domain or longer period, not a different test.

8.3 Recommendations

From these results we derive the following recommendations for comparing two models on spatio-temporal data.

1. *Define the estimand first.* Choose the validation design (random, spatial, temporal or spatio-temporal blocks, with buffers) to match the intended use of the model; random cross-validation answers an interpolation question.
2. *Never rely on procedures that treat observations as independent* (i.i.d. bootstrap, z- or t-tests on pooled observations, McNemar's test) or on the paired t-test with random folds. Under dependence, the corrected resampled t-test is not a remedy either.
3. *Use blocked cross-validation with a dependence-aware test.* Report DA-HAC together with the paired t-test on blocked folds; a conclusion that holds under both is more credible than one that holds under either. In our experiments, rejecting only when both tests reject reduced the error rate by 1–3 percentage points at a small cost in power (an exploratory, post-hoc observation that would deserve a dedicated study).
4. *Inspect the diagnostics.* Report the estimated ranges, the effective sample size and the range ratio. If the range ratio exceeds about 0.25, or if the effective sample size is a few dozen, treat any p-value as optimistic and avoid strong claims.
5. *Report intervals, not only p-values*, on the scale of the loss, as advocated for large data sets in this journal [33]: an improvement can be statistically real and practically negligible, or practically important and statistically unresolved.

8.4 Limitations and future work

Our simulation designs are stylised: separable Gaussian fields on regular grids, stationary dependence, and a single pair of learners. The working model (2) assumes additive fold effects and stationarity of the loss differential, which may fail when one model is better in some regions and worse in others; a non-stationary extension with spatially varying Δ is a natural next step. The training correlation ρ is set to the Nadeau–Bengio value rather than estimated, as it cannot be identified from a single cross-validation run [13]; repeated blocked cross-validation could provide information on it. We considered additive losses only; non-additive metrics such as the AUC or the F1-score would require linearisation. Finally, comparisons of more than two models raise multiplicity issues [3, 4] that combine with dependence and are left for future work.

9 Conclusion

Whether a machine learning model is "better" than another is a statistical claim, and on spatio-temporal data it depends on two choices that are rarely made explicit: the validation design, which defines what is estimated, and the inferential procedure, which defines how its uncertainty is assessed. We formalised the comparison of two models as inference on the mean of a dependent loss-differential field with correlated fold effects, showed why the paired t-test on fold scores fails under random cross-validation, and

proposed DA-HAC, a test that combines a variogram-calibrated product-kernel variance, a training-variance term and a Satterthwaite reference distribution, all computed from one blocked cross-validation run.

In controlled simulations and with gradient-boosting learners, conventional procedures declared false improvements up to sixteen times more often than their nominal level, and random cross-validation could halve the estimated gain of a genuinely better model. Blocked cross-validation combined with DA-HAC, or with the paired t-test on blocked folds, brought the error rate close to the nominal level under weak dependence and reduced it substantially under stronger dependence, without fully restoring it when the dependence range approaches the size of the study domain, a regime that DA-HAC's diagnostics help to detect. On three public data sets, the conclusion of four of the six comparisons depended on the validation design or on the test: an apparent superiority of XGBoost on hourly weather data vanished under temporal blocking, and two smaller gains were declared significant by independence-based procedures but not by DA-HAC, whose diagnostics showed that one of them could not be resolved with the available data. Large improvements, such as those of gradient boosting over linear baselines, were confirmed by every procedure.

Trustworthy machine learning requires not only accurate and explainable models but also trustworthy evidence that one model is better than another. We hope that the procedure, the diagnostics and the open-source implementation presented here will help practitioners in environmental, agricultural, epidemiological and mobility applications to report improvements that are statistically real.

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- **Conflict of interest.** The author declares no competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** All data sets are public and are redistributed, with their original licences, in the code repository, together with the raw outputs of all experiments.
- **Code availability.** The Python package `dacomp` and all scripts needed to reproduce the simulations, case studies, tables and figures are available at https://github.com/2zalab/Statistics-for-Data-Science-and-AI under the MIT licence.
- **Author contribution.** I.T. conceived the study, developed the methodology and software, carried out the experiments and wrote the manuscript.

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Table 5 Case studies: estimated loss differential $\hat{\Delta}$ (XGBoost minus competitor; negative favours XGBoost) with 95% confidence interval and p-value. R: random 5-fold CV; B: blocked 5-fold CV. Losses: log loss (NYC weather), squared error of $\log(1 + \text{count})$ (pedestrians), squared error of yield in $(\text{q/ha})^2$ (maize).

CV	Procedure	$\hat{\Delta}$ [95% CI]	p
NYC weather (rain, log loss)			
XGBoost vs random forest			
R	Paired t (folds)	-0.0112 [-0.0121, -0.0103]	<0.001
R	i.i.d. bootstrap	-0.0112 [-0.0131, -0.0096]	0.001
R	Corrected resampled t	-0.0115 [-0.0131, -0.0098]	<0.001
B	Paired t (folds)	0.0035 [-0.0082, 0.0152]	0.454
B	i.i.d. bootstrap	0.0036 [0.0005, 0.0067]	0.023
B	Block bootstrap	0.0036 [-0.0028, 0.0099]	0.279
B	Mixed model	0.0036 [-0.0033, 0.0105]	0.291
B	DM–HAC (time only)	0.0036 [-0.0007, 0.0079]	0.099
B	DA-HAC (proposed)	0.0036 [-0.0076, 0.0148]	0.499
XGBoost vs linear			
R	Paired t (folds)	-0.0413 [-0.0447, -0.0378]	<0.001
R	i.i.d. bootstrap	-0.0413 [-0.0445, -0.0377]	0.001
R	Corrected resampled t	-0.0413 [-0.0453, -0.0373]	<0.001
B	Paired t (folds)	0.0032 [-0.0075, 0.0139]	0.459
B	i.i.d. bootstrap	0.0028 [-0.0010, 0.0062]	0.119
B	Block bootstrap	0.0028 [-0.0046, 0.0116]	0.495
B	Mixed model	0.0028 [-0.0054, 0.0111]	0.487
B	DM–HAC (time only)	0.0028 [-0.0030, 0.0087]	0.344
B	DA-HAC (proposed)	0.0028 [-0.0052, 0.0109]	0.491
Melbourne pedestrians (log count, MSE)			
XGBoost vs random forest			
R	Paired t (folds)	0.0003 [-0.0024, 0.0030]	0.795
R	i.i.d. bootstrap	0.0003 [-0.0015, 0.0019]	0.757
R	Corrected resampled t	0.0003 [-0.0020, 0.0027]	0.782
B	Paired t (folds)	-0.0025 [-0.0047, -0.0003]	0.033
B	i.i.d. bootstrap	-0.0025 [-0.0043, -0.0007]	0.011
B	Block bootstrap	-0.0025 [-0.0090, 0.0027]	0.402
B	Mixed model	-0.0025 [-0.0086, 0.0036]	0.415
B	DM–HAC (time only)	-0.0025 [-0.0060, 0.0010]	0.156
B	DA-HAC (proposed)	-0.0025 [-0.0061, 0.0011]	0.172
XGBoost vs linear			
R	Paired t (folds)	-1.89 [-1.91, -1.88]	<0.001
R	i.i.d. bootstrap	-1.89 [-1.91, -1.87]	0.001
R	Corrected resampled t	-1.89 [-1.91, -1.87]	<0.001
B	Paired t (folds)	-1.85 [-1.90, -1.80]	<0.001
B	i.i.d. bootstrap	-1.85 [-1.87, -1.83]	0.001
B	Block bootstrap	-1.85 [-1.91, -1.78]	0.001
B	Mixed model	-1.85 [-1.91, -1.78]	<0.001
B	DM–HAC (time only)	-1.85 [-1.89, -1.81]	<0.001
B	DA-HAC (proposed)	-1.85 [-1.90, -1.80]	<0.001
Las Rosas maize (yield, MSE)			
XGBoost vs random forest			
R	Paired t (folds)	-2.82 [-4.64, -0.99]	0.013
R	i.i.d. bootstrap	-2.82 [-3.67, -1.81]	0.001
R	Corrected resampled t	-2.36 [-3.53, -1.19]	<0.001
B	Paired t (folds)	-20.9 [-70.4, 28.7]	0.308
B	i.i.d. bootstrap	-21.1 [-24.4, -18.1]	0.001
B	Block bootstrap	-21.1 [-47.0, -3.3]	0.043
B	Mixed model	-21.1 [-44.0, 1.8]	0.069
B	DA-HAC (proposed)	-21.1 [-88.3, 46.1]	0.449
XGBoost vs linear			
R	Paired t (folds)	-131.9 [-141.2, -122.7]	<0.001
R	i.i.d. bootstrap	-131.9 [-138.1, -126.0]	0.001
R	Corrected resampled t	-131.7 [-137.5, -125.9]	<0.001
B	Paired t (folds)	-160.0 [-193.0, -127.1]	<0.001
B	i.i.d. bootstrap	-160.0 [-167.7, -152.1]	0.001
B	Block bootstrap	-160.0 [-210.4, -112.3]	0.001
B	Mixed model	-160.0 [-210.0, -109.9]	<0.001
B	DA-HAC (proposed)	-160.0 [-210.3, -109.6]	<0.001

Table 6 Case studies: DA-HAC diagnostics on the blocked loss differentials: estimated practical ranges in space and time, effective sample size $\hat{n}_{\text{eff}}$, degrees of freedom $\hat{\nu}$ and training variance $\hat{\tau}^2$.

Data set	Comparison	$\hat{r}_S$	$\hat{r}_T$	$\hat{n}_{\text{eff}}$	$\hat{\nu}$	$\hat{\tau}^2$
NYC weather	XGBoost vs random forest	–	20 h	5143	12	3.37e-05
NYC weather	XGBoost vs linear	–	7 h	5276	1857	0.00e+00
Melbourne pedestrians	XGBoost vs random forest	–	5 h	17442	5008	0.00e+00
Melbourne pedestrians	XGBoost vs linear	–	38 h	8584	689	0.00e+00
Las Rosas maize	XGBoost vs random forest	153 m	–	154	5	1.35e+03
Las Rosas maize	XGBoost vs linear	171 m	–	91	20	0.00e+00

Table 7 Seed-null diagnostic: percentage of 30 pairs of XGBoost runs differing only in their random seed for which the procedure rejects equality at $\alpha = 0.05$. R: random CV; B: blocked CV; –: not available (no time axis).

	Random CV		Blocked CV					
Data set	Paired t	i.i.d. bs	Paired t	i.i.d. bs	Block bs	LMM	DM–HAC	DA-HAC
NYC weather	0	7	13	33	10	10	23	10
Melbourne pedestrians	0	43	7	67	0	0	33	3
Las Rosas maize	3	7	0	87	0	0	–	0